So far, you've been developing smart contracts and Dapps and deploying them on a test chain, such as the Remix IDE's JavaScript VM or the Ganache local test chain, to gain experience in a controlled environment. Armed with the knowledge of cryptography basics, you can now graduate to deploying smart contracts on a public chain. Ropsten is a public test network that implements the Ethereum blockchain protocol, but with mock ether. Ropsten is ideal for experimenting with your deployment after completing initial tests in Remix and other test networks. Before you deploy on Ropsten, you'll need a few items to set up the environment:

- A wallet for managing accounts and signing transactions. You can use MetaMask to manage your Ropsten accounts and their (test) ether balance.
- A method for populating this wallet with a deterministic set of test account addresses.
- A Ropsten faucet for depositing test ether into the accounts for Tx execution and ether transfer among peer participants.
- The Remix IDE's injected web3 environment for supporting the Ropsten accounts through MetaMask and interacting with the smart contract through its user interface.
- A smart contract ready to be deployed on the Ropsten network. You'll use the Counter smart contract (section 5.2.6) for this initial deployment.

5.2.4 Using the private key in mnemonic form

The 160-bit account addresses are cryptographically generated from the 256-bit private-public key pair. You need this private key every time you want to generate/recall your account addresses. It is simply impossible for you to remember this private key. Instead, a mnemonic is used to represent the private key.

> **NOTE** *BIP39* (http://mng.bz/awoJ) stands for *Bitcoin Improvement Protocol 39*, which was developed to define a method to use a mnemonic to represent a private key. The